Do Now · retrieval

Processors, scheduling and LMC practice

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does LMC STA do?

Q2

What is the role of MAR?

Q3

What is a time quantum?

Q4

Full adder 1+1+0: sum and carry bits?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — Trace the LMC accumulator and mailbox separately: INP overwrites ACC, while STA copies to storage.

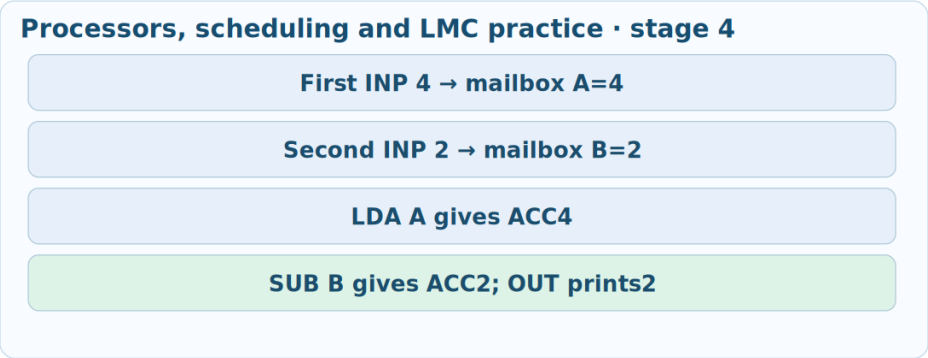
Core idea 2

Apply — Schedule independent CPU jobs with an explicit arrival/time rule and label every Gantt boundary.

Core idea 3

Evaluate — Distinguish processor speed, memory cache and interrupt/OS services when explaining performance.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

ACC — Accumulator register.

FCFS — First Come First Served.

SJF — Shortest Job First.

pipeline — Overlap instruction stages.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

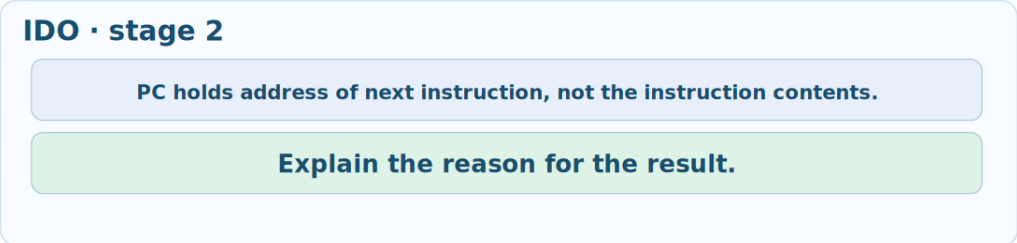
Problem

Trace sequential instruction fetch when PC=90.

Follow the worked method

1. Predict first — PC holds address of next instruction, not the instruction contents.
2. Explain and check — Explain the reason for the result.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
Arrival-order service makes B wait three CPU units.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Using LMC code above, trace ACC after each input, LDA, SUB and OUT. [6]

Question 2

Processes A (burst 4), B (burst 2) and C (burst 1) all arrive at t=0. For FCFS, use the stated arrival-list order A, B, C to break ties; for non-preemptive SJF choose the shortest available job. Compare execution orders and mean waiting times. [6]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate clock speed, cache and pipelining for a task with heavy memory latency and branches. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

LMC STA X loads ACC from mailbox X.

Use key vocabulary from memory

1. Explain ACC in your own words.

2. Explain FCFS in your own words.

3. Explain SJF in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.